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Vectors and linear maps

*A useful definition should help us compute,
but it should also help us see.*

A guiding principle for this example

1.0 INTRODUCTION

This example shows the fixed left sidebar and the book's section-based numbering. It uses Computer Modern for text and mathematics. The sample text and drawings are original.

Linear algebra studies operations that preserve addition and scalar multiplication. These operations occur whenever several quantities are combined without interaction terms: adding displacements, changing coordinates, or superposing solutions of a homogeneous equation.

The point of introducing an abstract definition is to recognize the same structure in different settings. A column of numbers, a polynomial, and a continuous function can all be vectors. Their concrete descriptions differ, but many of the arguments we make about them are identical.

We move between coordinates and geometry. The main column develops the argument; the margin supplies interpretations and reminders.

1.1 VECTORS AND LENGTH

A point describes a location; a vector describes a displacement. After choosing an origin, we may represent both by the same list of coordinates, but it is useful to remember their different roles. We can add two displacements without making any further choice.

Definition 1.1.1 (Euclidean inner product). For $u, v \in \mathbb{R}^n$, their *inner product* is

$$\langle u, v \rangle = \sum_{j=1}^n u_j v_j. \quad 1.1.1$$

The associated length is $\|v\| = \sqrt{\langle v, v \rangle}$.

For a nonzero vector, length and direction carry different information. Dividing by the length produces a unit vector with the same direction.

The inner product is linear in each variable and symmetric in its two arguments. It is also positive: $\langle v, v \rangle \geq 0$, with equality precisely when every coordinate of v vanishes. These observations follow directly from the coordinate formula.

Example 1.1.2 (A unit vector). The vector $v = (3, 4)$ has length 5. Therefore $\frac{1}{5}v = (\frac{3}{5}, \frac{4}{5})$ has length 1. $\triangle$

A geometric interpretation

In the plane, the coordinate formula for length is the Pythagorean theorem. Figure 1.1.1 separates a displacement into horizontal and vertical parts. The same formula in higher dimensions adds the squares of all the coordinate displacements.

Angles can also be recovered from the inner product. For nonzero vectors u and v , the cosine of the angle between them is

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}. \quad 1.1.2$$

The following inequality ensures that the quotient lies in $[-1, 1]$. It will also give us the triangle inequality, which expresses the fact that traveling directly is no longer than traveling in two stages.

Theorem 1.1.3 (Cauchy–Schwarz inequality). *For all $u, v \in \mathbb{R}^n$,*

$$|\langle u, v \rangle| \leq \|u\| \|v\|. \quad 1.1.3$$

Proof. The result is immediate if $v = 0$. Otherwise, for every real t ,

$$\begin{aligned} 0 &\leq \|u - tv\|^2 \\ &= \|u\|^2 - 2t\langle u, v \rangle + t^2\|v\|^2. \end{aligned}$$

Set $t = \langle u, v \rangle / \|v\|^2$. Substitution gives $0 \leq \|u\|^2 - \langle u, v \rangle^2 / \|v\|^2$, which is the desired inequality after multiplication and taking square roots. $\square$

Equality in Cauchy–Schwarz occurs when the vectors are linearly dependent. In the proof, it occurs when the residual vector $u - tv$ can be made zero.

Corollary 1.1.4 (Triangle inequality). *For all $u, v \in \mathbb{R}^n$, $\|u + v\| \leq \|u\| + \|v\|$.*

Proof. Expand the square of the left side and apply Theorem 1.1.3:

$$\|u + v\|^2 \leq \|u\|^2 + 2\|u\| \|v\| + \|v\|^2 = (\|u\| + \|v\|)^2.$$

Both sides are nonnegative, so we may take square roots. $\square$

Remark. A coordinate formula can suggest a geometric statement; a geometric picture can suggest the right algebraic proof. Neither viewpoint needs to replace the other.

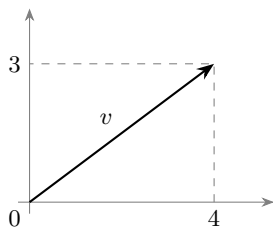


FIGURE 1.1.1. The vector $v = (4, 3)$ and its coordinate displacements.

1.2 LINEAR MAPS

The same definition applies to arbitrary real vector spaces. Coordinates are convenient for calculation, but they are not part of the underlying idea of linearity.

The operations of addition and scalar multiplication determine what a linear map must preserve. One compact identity expresses both requirements.

Definition 1.2.1 (Linear map). A map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *linear* if

$$T(au + bv) = aT(u) + bT(v) \quad 1.2.1$$

for every $u, v \in \mathbb{R}^n$ and every $a, b \in \mathbb{R}$.

Example 1.2.2 (A shear). The map $S(x, y) = (x + y, y)$ is linear. It leaves the horizontal axis fixed and shifts every other horizontal line by an amount proportional to its height. Its matrix in the standard basis is

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad S \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}. \quad 1.2.2$$

△

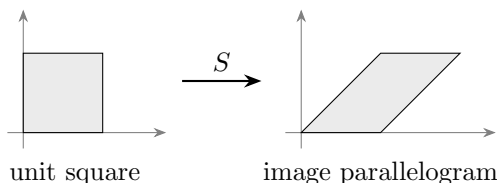


FIGURE 1.2.1. A shear changes angles while preserving parallel lines and area.

What happens to the origin?

A translation $x \mapsto x + c$, with $c \neq 0$, is not linear: it moves the origin. It is an affine map. This distinction matters even when the graph looks like a straight line.

Proposition 1.2.3. Every linear map sends the zero vector to the zero vector.

Proof. Since $0 = 0 + 0$, linearity gives $T(0) = T(0) + T(0)$. Subtracting $T(0)$ proves the claim. □

A useful test. Before checking the full definition, evaluate the proposed map at zero. If the result is nonzero, the map cannot be linear.

Matrices encode the images of a basis

Let $e_1, \dots, e_n$ be the standard basis. Every vector has the expansion $v = \sum_j v_j e_j$. If T is linear, we therefore have

$$T(v) = \sum_{j=1}^n v_j T(e_j). \quad 1.2.3$$

The vectors $T(e_j)$ completely determine the map. Arrange them as the columns of a matrix A ; equation 1.2.3 then says $T(v) = Av$.

This is an automatically placed note. It can move downward to avoid another automatic note. It is useful when exact vertical alignment is less important than keeping a sequence of notes apart.

Theorem 1.2.4 (Basis criterion). *Two linear maps $T, U: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are equal if and only if $T(e_j) = U(e_j)$ for every standard basis vector e_j .*

Proof. One direction is immediate. For the other, expand an arbitrary vector in the standard basis and use linearity:

$$T(v) = \sum_j v_j T(e_j) = \sum_j v_j U(e_j) = U(v).$$

Thus the two maps agree at every vector. □

Reading a matrix by columns. The j th column answers a concrete question: where does the map send the j th basis vector? This interpretation is often easier to remember than a formula involving two indices.

Common pitfall. Preserving lengths is a much stronger requirement than being linear. The shear in Example 1.2.2 is linear, but it sends $(0, 1)$ to $(1, 1)$, changing the length from 1 to $\sqrt{2}$.

EXERCISES FOR SECTION 1.2

1.2.1 Show that every linear map satisfies $T(-v) = -T(v)$.

1.2.2 Find the matrix of $T(x, y) = (2x - y, x + 3y)$. What are the images of the two standard basis vectors?

1.2.3 (Composition) Prove that the composition of two linear maps is linear. If their matrices are A and B , in that order, what is the matrix of the composition?

1.2.4 Give a map $f: \mathbb{R} \rightarrow \mathbb{R}$ that fixes zero but is not linear.

2

Orthogonal decomposition

*Choose the component that explains the direction,
then study what remains.*

A second viewpoint

2.0 INTRODUCTION

Many calculations become simpler after splitting a vector into two perpendicular pieces. One piece lies in a chosen direction; the other records the residual. This is the basic geometry behind projection and least-squares approximation.

2.1 PROJECTION ONTO A LINE

Fix a nonzero vector u . We seek a scalar a for which $v - au$ is perpendicular to u . The condition $\langle v - au, u \rangle = 0$ gives a unique answer.

The denominator is $\|u\|^2$, so the expression is defined only when $u \neq 0$. Multiplying u by a nonzero scalar does not change the projection map.

Definition 2.1.1 (Orthogonal projection). The projection of v onto the line spanned by $u \neq 0$ is

$$P_u(v) = \frac{\langle v, u \rangle}{\langle u, u \rangle} u. \quad 2.1.1$$

The map P_u is linear. Moreover, $P_u(v)$ lies on the chosen line and $v - P_u(v)$ is perpendicular to it. These properties characterize the decomposition uniquely.

Theorem 2.1.2 (Best approximation). Among all vectors w on the line spanned by u , the vector $P_u(v)$ uniquely minimizes $\|v - w\|$.

Proof. Write $v - w = (v - P_u(v)) + (P_u(v) - w)$. The two summands are perpendicular, so

$$\|v - w\|^2 = \|v - P_u(v)\|^2 + \|P_u(v) - w\|^2.$$

The final term is nonnegative and vanishes exactly when $w = P_u(v)$. $\square$

Remark. This is a geometric optimization argument: perpendicularity converts the quantity being minimized into a fixed term plus a nonnegative term.

Seeing the residual

The following drawing occupies both columns. A wide figure is useful when the relation between several objects would be cramped in the main column. Leave the corresponding margin area free of notes.

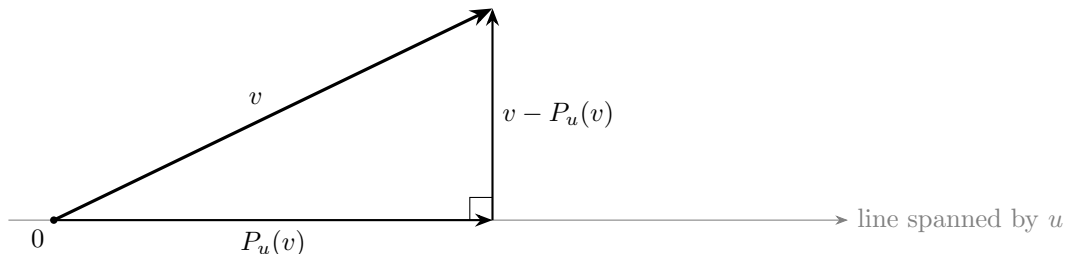


FIGURE 2.1.1. An orthogonal decomposition into a component on a line and a residual perpendicular to it. The caption and diagram use the full width.

Applying the projection twice has the same effect as applying it once:

$$P_u^2 = P_u. \quad 2.1.2$$

A map with this property is called *idempotent*. Once a vector has landed on the chosen line, a further projection leaves it unchanged.

A numerical example

¹ A margin note can also carry a number. The reference mainly uses unnumbered notes. Ordinary footnotes remain available.

¹ Let $u = (1, 1)$ and $v = (2, 0)$. Then $P_u(v) = (1, 1)$, and the residual is $(1, -1)$. Their inner product is zero, as expected. The squared length of v is 4; the squared lengths of its two perpendicular components are 2 and 2.

Definition 2.1.3 (Idempotent map). A linear map $P: V \rightarrow V$ is *idempotent* if $P^2 = P$.

EXERCISES FOR SECTION 2.1

2.1.1 Verify directly from equation 2.1.1 that $P_u^2 = P_u$.

2.1.2 Find the matrix of the orthogonal projection onto the line spanned by $(1, 2)$. Check that the matrix is symmetric and idempotent.